

Saeid Samadi — Robotics Laboratory Manager

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Profile

With over six years in the field of robotics and artificial intelligence, I've extensively worked with humanoid, industrial, and bio-inspired robots. My expertise spans motion planning, whole-body control, and innovative robot design. From research to leading a robotics laboratory, I've collaborated with international teams and contributed to key publications. Committed to continuous learning, I aim to drive innovation in robotics.

Education

University of Montpellier

Ph.D. in Robotics and AI

2018–2022

Thesis: "Humanoid robots balance in multi-contact modes settings"

University of Tehran

M.Sc. in Mechanical Engineering

2016–2018

Thesis: "Designing an Online Walking Algorithm of Humanoid Robot on Rough Surfaces with Different Heights"

University of Tabriz

B.Sc. in Mechanical Engineering

2012–2016

Project: "Free-convection analysis of an elliptic cylinder"

Professional Experience

University of Edinburgh

Laboratory Manager (Permanent Position)

June 2023–Present

- Overseeing the operations and research activities of the robotics laboratory.
- Collaborating with faculty and students to drive forward innovative projects.
- Ensuring the maintenance and upgrade of laboratory equipment and tools.
- Organizing workshops, seminars, and collaborative events with industry partners.

University of Edinburgh

Research Associate

Mar 2022–June 2023

- Post-doctoral researcher.
- Research in Statistical Machine Learning and Motor Control.
- Collaboration on the HARMONY EU project.
- Design and implementation of robot control interfaces.

LIRMM, Montpellier

Researcher & Ph.D. Student

2018–Feb 2022

- Real-time balance of humanoid robots in multi-contact modes.
- Experiments with FOGALE sensors for direct robot adjustments.

University of Tehran

Teaching Assistant

2017–2018

- Assisted in Smart Structure course.
- and Fuzzy Control courses.
- Mentored graduate students on project work.

Skills

Programming: C++, Python, MATLAB

Tools: ROS, Gazebo, Choreonoid, SolidWorks, GIT, Docker, YAML, CMake

Hardware: CNC machine, 3D printing

Core Competencies: Team Leadership, Analytical Thinking, Clear Communication, Team Collaboration

Languages

Fluent: English, Persian, Azeri, Turkish

Intermediate: French, Arabic

Basic: German

Publications

Under Review.....

- **W. Du, R. Long, J. Moura, J. Wang, S. Samadi, S. Vijayakumar**, "An Efficient Representation of Whole-body Model Predictive Control for Online Compliant Dual-arm Mobile Manipulation", *IEEE Transactions on Robotics (T-RO)*.

In Preparation.....

- **S. Samadi, P. Gergondet, S. Vijayakumar, and A. Kheddar**, "Multi-Modal Whole-Body Balance Control for Humanoids in Real-Time Multi-Contact Scenarios", *IEEE Transactions on Robotics (T-RO)*.

Published Works.....

For a comprehensive list of my published works, please refer to my Google Scholar Profile [Link].

Selected Projects

- **Talos Humanoid Robot (2022 - Present, SLMC):** Developed control interfaces, balance strategies, and footstep planning algorithms. *Technologies:* C++, Gazebo, ROS, mc_rtc.
- **HARMONY EU Horizon Project (April 2022 - June 2023):** Contributed to a project enhancing healthcare via assistive robotic manipulation, developing navigation and control for a wheeled robot. *Technologies:* [C++, Python, ROS, Gazebo]
- **HRP-4 Humanoid Robot (2018 - 2022, LIRMM):** Worked on real-time balance in multi-contact modes and conducted experiments with FOGALE sensors. *Technologies:* C++, Python, Choreonoid, ROS.
- **AI Projects (2017 - 2018, University of Tehran):** Implemented game strategies and neural fuzzy systems. *Technologies:* Matlab, LabView.
- **Smart Structures (2016 - 2017, University of Tehran):** Worked on SMA-actuated controls and system identification. *Technologies:* Matlab, LabView.

Honor Awards and Services

- **Reviewer for International Conferences and Journals (2018 - Present):** Journals and conferences including TRO, RAL, ICRA, IROS, and Humanoids.
- **M.Sc. Fellowship by MSRT, Iran (2016):** Ranked 347th out of 15,000 in the entrance exam.
- **M.Sc. Fellowship at University of Tabriz (2016):** Among top 10% based on undergraduate GPA.
- **B.Sc. Fellowship by MSRT, Iran (2012):** Top 0.7% among 260,055 in the entrance exam.
- **Educational Acceleration (2007):** Completed second grade of Middle School early.